



# First report of *Monalonion velezangeli* (Hemiptera: Miridae) damaging avocado (*Persea americana*) in Peru

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**Abstract** Avocado (*Persea americana* Mill.) is an economically important fruit crop whose expansion has been accompanied by the emergence of new insect pests affecting yield and fruit quality. During field surveys conducted in avocado plantations in the Amazonas region of northern Peru, severe damage associated with a mirid bug was observed on young fruits and vegetative shoots. This study aimed to identify the causal insect and characterize the associated damage through morphological and molecular analyses. Adult and nymph specimens were collected from five plantations and identified using diagnostic morphological traits and taxonomic keys. Molecular confirmation was performed using mitochondrial cytochrome c oxidase subunit I (COI) sequences and phylogenetic analysis based on the Maximum

Likelihood method. Both morphological and molecular evidence confirmed the species as *Monalonion velezangeli*. Field observations showed that the insect primarily attacks young fruits and shoots, causing necrotic lesions and punctures that reduce fruit commercial quality. This study represents the first confirmed report of *M. velezangeli* affecting avocado in Peru.

**Keywords** Avocado pest · Emerging pest · *Persea americana* · *Monalonion* · Peru

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## Introduction

Avocado (*Persea americana* Mill.) cultivation has become one of the most economically dynamic fruit crops worldwide due to its high nutritional value and growing demand in international markets, as well as its use in the food, cosmetic, and pharmaceutical industries (Cheptoo et al., 2025; Dreher & Davenport, 2013; Fernando et al., 2025). In 2024, global production reached 11.2 million tons, with Mexico as the leading producer and Peru ranking fourth and as the world's second-largest exporter (FAO, 2026). In Peru, avocado cultivation is characterized by a strong presence of smallholder farmers, with approximately 97% managing less than 5 ha, making it a key activity for the rural economy (MIDAGRI, 2025; PROHASS, 2022).

However, the expansion of the crop has facilitated the emergence of pests that affect fruit productivity and quality, including phytophagous insects from the orders Lepidoptera, Coleoptera, and Hemiptera (Carabali-Banguero et al., 2018; Hoddle & Brown, 2010). Among the latter, several species of the family Miridae are considered emerging pests in tropical crops (Arellano & Vergara, 2016; Wheeler & Henry, 2004). Notably, *Monalonion velezangeli* has been recognized as an economically important pest of avocado (*Persea americana*) and coffee (*Coffea arabica*) (Noboa et al., 2024; Torres Jaimes et al., 2012). The species was initially reported in Colombia affecting avocado and later confirmed in Ecuador, evidencing its expansion in Andean production areas (Noboa et al., 2024). This species is a polyphagous herbivore considered a relevant pest for various tropical crops due to its ability to feed on multiple host plants from different botanical families, which gives it notable ecological plasticity (Gamboa et al., 2020; Giraldo-Jaramillo et al., 2010). Damage caused by both nymphs and adults includes necrosis on shoots, flowers, and young fruits, significantly reducing commercial quality and generating economic losses in intensive production systems (Giraldo-Jaramillo et al., 2010; Noboa et al., 2024).

However, in Peru, information on its presence and distribution remains limited. Therefore, the present study aimed to report, for the first time, the presence of *M. velezangeli* in avocado plantations in the Peruvian Amazon through morphological and molecular identification, as well as to describe the damage observed in the field, providing scientific evidence

to support the monitoring and management of this emerging pest.

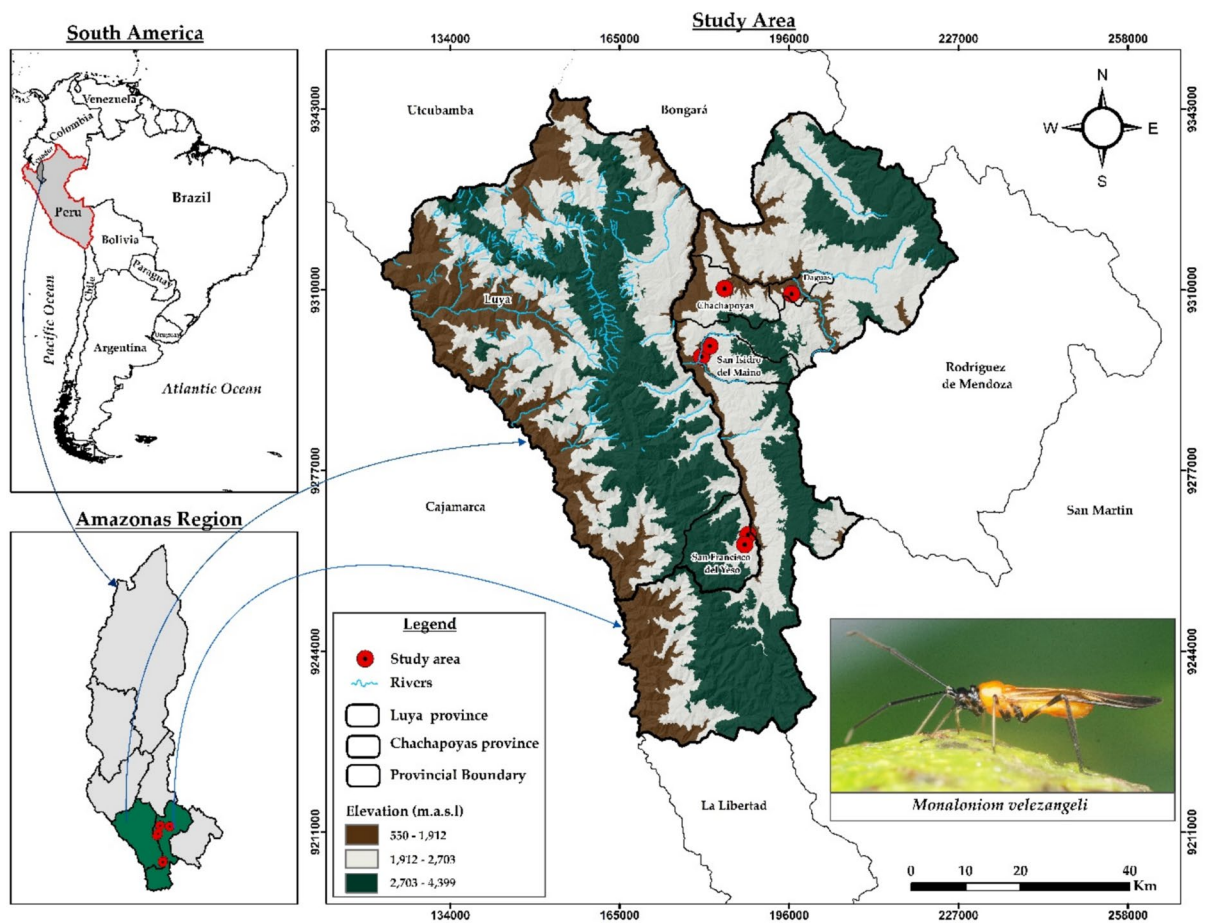
## Materials and methods

### Area studied and sample collection

Sampling was conducted in five avocado plantations across different production areas in Chachapoyas province, Amazonas, Peru, representing a range of altitudes (2,218–2,410 m a.s.l.), plantation ages (4–6 years), varieties (Fuerte and Hass), and planting densities (300–400 trees/ha). Plantations 1 and 2 were located in San Francisco del Yeso, plantations 3 and 5 in San Isidro de Mayno and San Francisco de Daguas, respectively, and plantation 4 consisted of a single Fuerte tree on the UNTRM university campus in the peri-urban area of Chachapoyas Fig. 1. The study sites are characterized by the typical dry montane scrubland vegetation of the eastern Andes, with a semi-humid climate characteristic of low, sheltered inter-Andean valleys (Mestanza et al., 2018). Temperatures range from 10 °C (June–August) to 22 °C (November–December), with annual seasonal precipitation of 800–1400 mm and relative humidity of 78%–85% (Rascón et al., 2022). After collection, 40 adults and 30 nymphs were transported to the Plant Health Research Laboratory at UNTRM. Part of the specimens were preserved in 30-mL vials containing 70% ethanol, whereas others were kept without preservative for subsequent morphological and molecular analyses.

### Identification and confirmation of damage caused by the avocado bug

Initial damage to avocado plants (young shoots and fruits) was observed in April 2025. Adult specimens were collected from affected plantations and confined with healthy shoots and fruits in 30×30×30 cm rearing cages to confirm pest activity (Fig. 2G, H). Damage to fruits and young stems was monitored and recorded 7 and 15 days after confinement, including perforations and other symptoms, which were documented photographically. A total of 15 cages were installed, three per plantation across the sampled districts.



**Fig. 1** Location of the *Monaloniom velezensis* sampling sites in the provinces of Luya and Chachapoyas, representative avocado production areas in the Amazonas region, Peru

### Morphological Identification

For morphological identification, a total of 40 adult and 30 nymph specimens at different developmental stages (8 adults and 6 nymphs per district) were collected. Specimens were preserved in 30 mL vials containing 70% ethanol. Identification was carried out using specific taxonomic keys (Carvalho & Costa, 1989) and supported by previous descriptions Leiva-Espinoza et al. (2026) and Noboa et al. (2024). Male genitalia were also examined following established preparation methods (Hardwick, 1950; Noboa et al., 2024). Photographic documentation of diagnostic characters was performed using a Sony Alpha IV camera with a 96 mm macro lens and a 2.0–5.0×Astrhori macro-objective, coupled with a Nikon SMZ1270 stereomicroscope. The identified

specimens were deposited in the entomology museum of the Plant Health Laboratory at Toribio Rodríguez de Mendoza National University in Amazonas.

### Molecular identification

#### *DNA Extraction, Amplification, and Sequence Alignment*

Total genomic DNA was extracted from pair of legs of adult insect samples, using the NucleoSpin® DNA Insect kit (Macherey–Nagel, Düren, Germany) following the manufacturer's protocol. DNA quality and concentration were determined using an Eppendorf BioSpectrometer® kinetic analyzer (Germany).

PCR amplification was performed in a total reaction volume of 25 µL containing 12.5 µL of Promega



**Fig. 2** Adults of *M. velezensis* causing damage to fruits (A); fruit with visible dark spots and white exudates (B); damage on young branches (C, D, E, F); in situ rearing cage (G); adult feeding on branches and young shoots (J, K)

GoTaq® PCR Master Mix (2×), 1.25 µL of forward primer (10 µM), 1.25 µL of reverse primer (10 µM), 5 µL of nuclease-free water, and 5 µL of DNA template. The mitochondrial cytochrome c oxidase subunit I (COI) gene was amplified following the thermal cycling protocol described by (Roca-Cusachs et al., 2022). Amplified products were sequenced using the Sanger method at SANGON BIOTECH (Shanghai).

### Phylogenetic analysis

COI gene sequences were edited using Sequencher v.5.4.6 (Gene Codes, Ann Arbor, Michigan, USA). The sequences were subsequently aligned using MEGA XII (Kumar et al., 2018). For first identification were performed in the NCBI (National Center for Biotechnology Information) database using the BLASTn tool, allowing comparison with previously reported sequences.

Phylogenetic analysis was performed using the Maximum Likelihood (ML) method implemented in RAxML-HPC BlackBox through the CIPRES Science Gateway platform (<https://www.phylo.org/>). The GTR+I+G substitution model was applied and node support was assessed with 1000 bootstrap replicates (Edler et al., 2021; Leiva-Espinoza et al., 2026).

## Results

### Characterization of feeding damage

Figure 2 illustrates the damage observed on avocado fruits and shoots, where both adults and nymphs were recorded at different developmental stages (Fig. 2A–C). The eggs were creamy white at the of oviposition, turning reddish-yellow before hatching (Figure 3I). They were located at the base of lateral buds on young branches, either singly or in pairs (Fig. 3G). Their presence was recognized by two white, filamentous projections, corresponding to respiratory ducts (Fig. 3H), which protruded above the plant tissue (Londoño Zuluaga, 2020).

Fruits showed localized depressions in the pericarp associated with dark necrotic lesions at feeding puncture sites, with lesion diameters ranging from 2 to 7 mm. As tissue deterioration progressed, these lesions frequently coalesced, forming larger irregular necrotic areas (Fig. 2A–B). In addition, white

crystalline exudates were observed around the puncture sites (Fig. 2B). These exudates are likely associated with the release and oxidation of perseitol, a characteristic polyol of avocado tissues, triggered by insect feeding activity (Noboa et al., 2026). After one week, lesions expanded into extensive necrotic patches that compromised fruit quality.

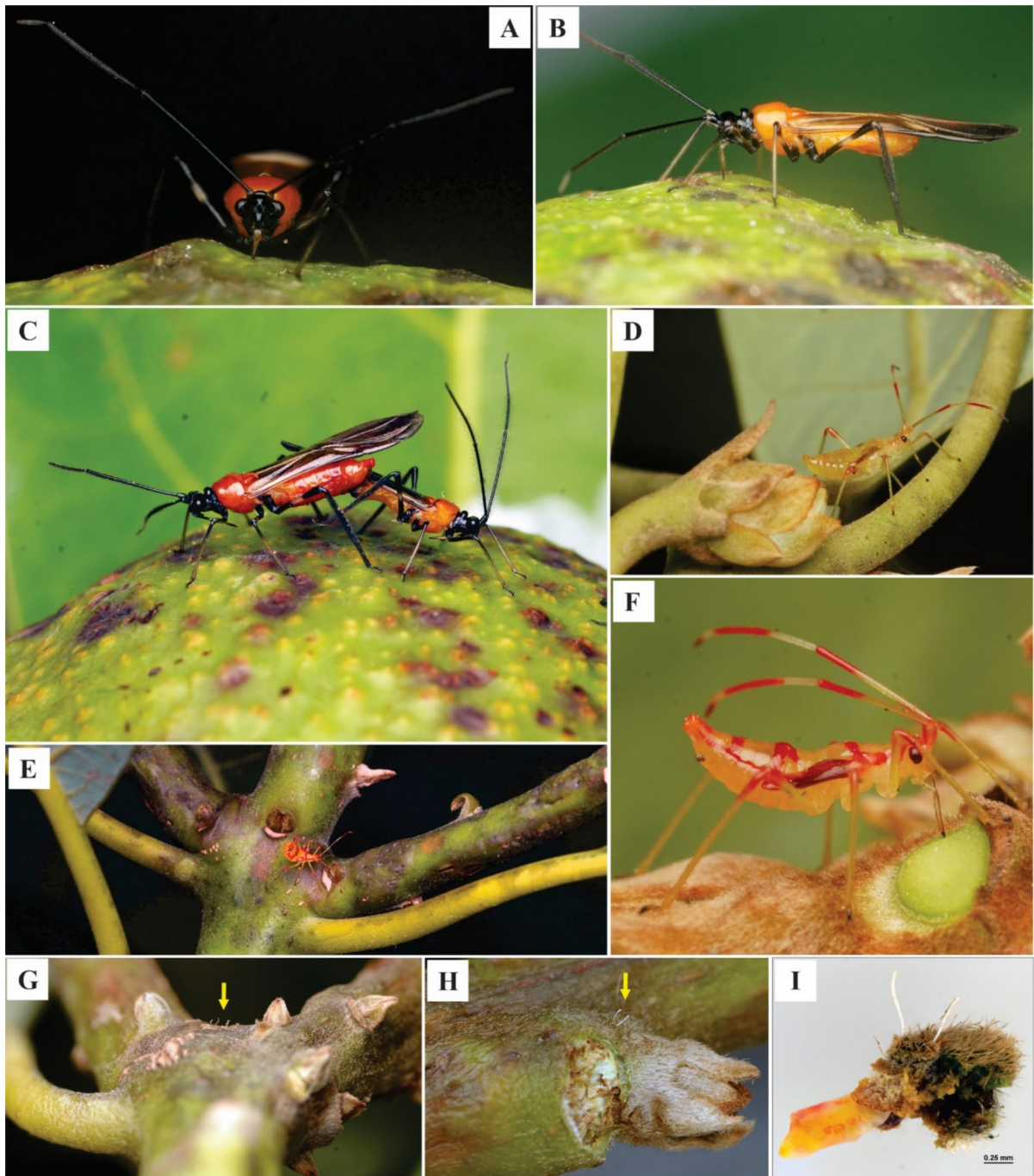
Damage on young branches initially appeared as dark feeding punctures resembling small welts, approximately 1–2 mm in diameter (Fig. 2D). After approximately one week, the affected branches exhibited superficial tissue deterioration and the release of crystalline exudates resembling sugar-like droplets (Fig. 2E–F). Yellowing and subsequent death of leaves, shoots, and young branches were observed, along with partial defoliation of the plant (Fig. 2C).

*M. velezangeli* showed a clear preference for young fruits, with damage recorded mainly on fruits measuring 1.5–5.5 cm in diameter, whereas no damage was observed on fruits close to maturity. On average, a single adult produced up to 50 feeding punctures per fruit.

### Morphological characterization

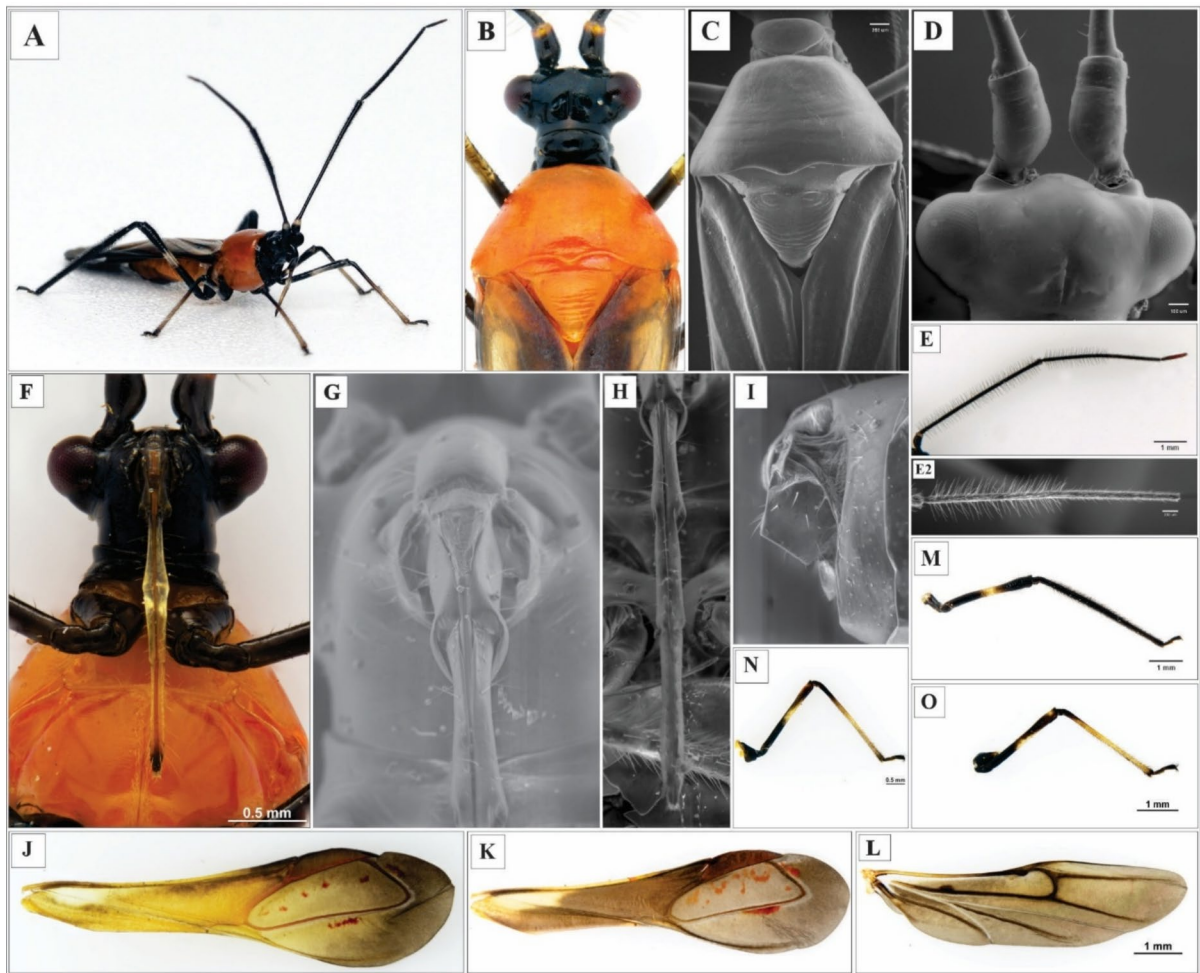
Adults exhibited coloration ranging from orange to black. Males showed a more slender body and a tendency toward orange coloration on the thorax and abdomen (Fig. 3B), whereas females displayed a more robust body with a tendency toward reddish coloration (Fig. 3A). The head was shiny black (Figs. 4B, F). Antennae were covered with black setae and composed of four segments, with the first and fourth segments visibly shorter (Fig. 4E, E2). The rostrum consisted of four segments and was inserted posterior to the first pair of legs (Fig. 4B, G, H, F). In addition, variation in the coloration of the scutellum and pronotum was observed.

The hind legs exhibited coloration ranging from brown to black, with the femora slightly thickened distally (Fig. 4M–O). A distinct white to creamy band was present in the median region of each femur, whereas the tibiae were slender and densely covered with setae (Fig. 4M). Adults possessed hemelytron-type forewings with coloration varying from ochre to dark brown or black. The corium typically presented four to five reddish spots within the discal cell along the main vein near the cuneus (Fig. 4J). In some individuals, these spots appeared irregular in shape and



**Fig. 3** In vivo view of *Monalonion velezangeli* individuals feeding on fruit. Female (A) and male (B); mating on damaged fruit (C); second nymphal stage (D); fourth nymphal stage (E,

F); egg-laying in groups of 2 (G); white filamentous projections (H); egg close to hatching (I)



**Fig. 4** Morphological features of *M. velezangeli* (A); pronotum and scutellum (B, C); posterior view of the head and first antennal segment (D); antenna showing its four segments (E); terminal antennal segment (E2); rostrum of *M. velezangeli* showing its four segments (F, G, H); terminal segment of the

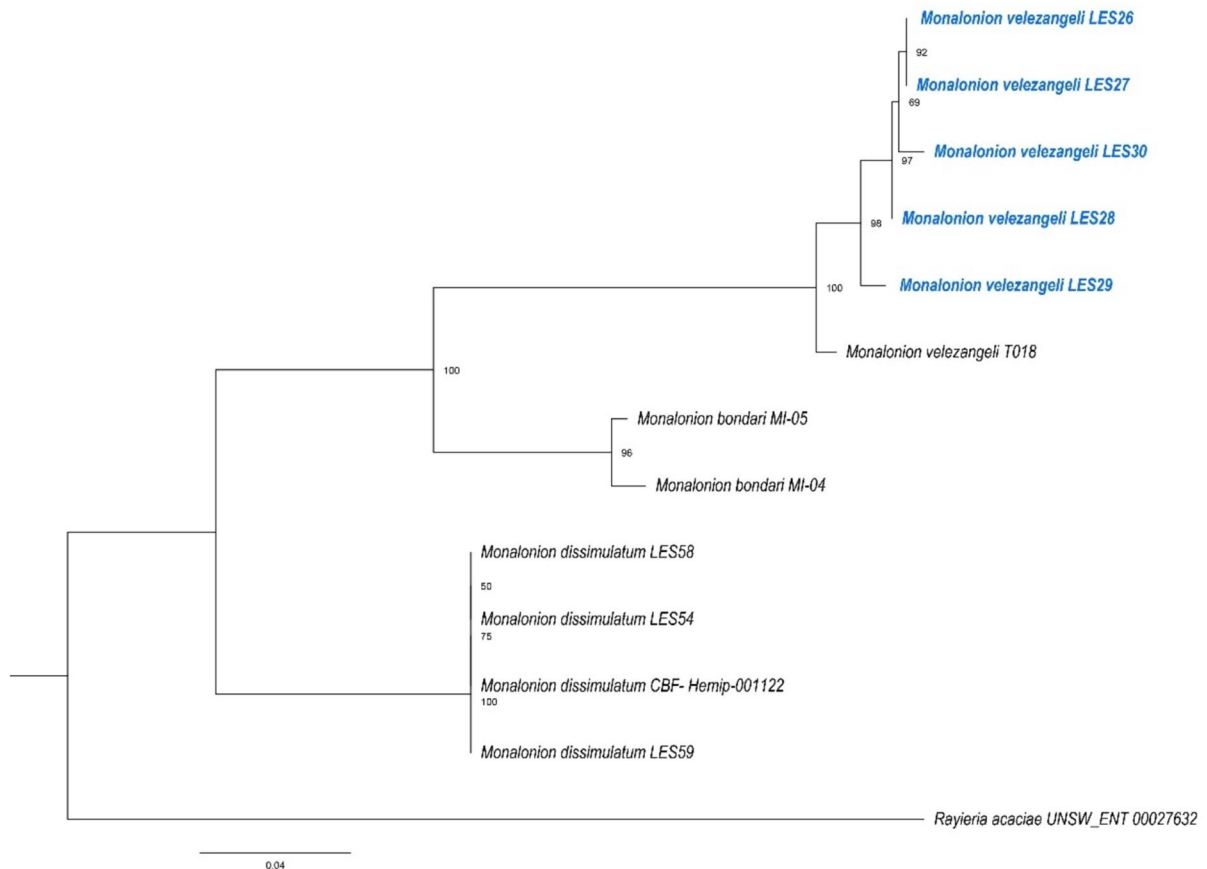
female abdomen (I); forewings with variable coloration in the corium and cuneus, showing distribution of internal and marginal spots on the membrane (J, K); hindwing (L); hind (M), mid (N), and fore (O) legs, all displaying creamy-white bands in the middle of the femora

distribution within the cell (Fig. 4K), and additional reddish markings were occasionally observed adjacent to the longitudinal veins of the corium (Fig. 4J, K).

Nymphs of *Monalonion velezangeli* pass through five developmental instars (Bolaños Muñoz, 2017). Body coloration ranged from yellow to reddish, with red spots distributed across the head, thorax, and abdomen. These markings became progressively more intense from the third instar onward, forming distinct red bands along the body segments (Fig. 3D–F). Nymphal antennae consisted of four

clearly defined segments with alternating reddish and yellowish coloration (Fig. 3F). The eyes were dark red, and additional reddish spots were present on the distal femur and basal tibia (Fig. 3D–F).

The male genitalia exhibited diagnostic characteristics typical of the genus *Monalonion*, including an aedeagus with a well-developed membranous portion and an elongated basal plate. The parameres were asymmetric, with the left paramere larger and more developed, whereas the right paramere was smaller, simple, slightly curved, and terminating in a rounded apex.



**Fig. 5** Phylogenetic tree based on COI gene sequences showing relationships among species of the genus *Monalonion*. Sequences obtained in this study (highlighted in blue) cluster

within the *Monalonion vellezangeli* clade. Node values represent bootstrap percentages. *Rayieria acaciae* was used as the outgroup

## Molecular Identification

Phylogenetic analysis based on mitochondrial COI sequences revealed a clear separation among species of the genus *Monalonion*. All sequences obtained in this study (LES26, LES27, LES28, LES29, and LES30) clustered within a monophyletic clade (high statistical support, bootstrap=100) together with the reference sequence PQ046647 (Voucher T018) corresponding to *Monalonion vellezangeli*, with high bootstrap support values (98–100), confirming their species-level taxonomic identity.

Within the clade, LES26 and LES27 grouped with 92% bootstrap support, while LES28, LES29, and LES30 (Fig. 5) showed support values between 97%–100%, indicating low intraspecific genetic divergence and high molecular consistency among

the analyzed individuals. The *M. vellezangeli* clade was recovered as a sister group to *Monalonion bondari*, represented by two isolates (OR961530 and OR961531) with 96% bootstrap support. In contrast, sequences assigned to *Monalonion dissimulatum* formed an independent clade, evidencing clear genetic divergence from the other analyzed species.

## Discussion

The damage observed on avocado fruits and young shoots evidences the phytophagous behavior of a piercing–sucking insect of the family Miridae, with a preference for young tissues, which facilitates the penetration of the mouthparts and the ingestion of cellular sap (Wheeler, 2001). The absence of damage in

fruits close to maturity is associated with the greater hardness of the tissues. Likewise, the simultaneous presence of nymphs and adults on fruits and branches indicates that avocado acts both as a food source and as a reproductive host, favoring the population establishment of the insect, a pattern consistent with the biological cycle described for species of the genus *Monalonion* (Giraldo-Jaramillo et al., 2010; Wheeler, 2001).

Initial lesions consist of small punctures that evolve into necrotic spots and depressions on the fruit, reflecting a progressive process of tissue deterioration induced by insect feeding (Ramírez-Gil et al., 2019). At first glance, fruit lesions can be confused with those caused by *Colletotrichum* spp., which appear as dark, circular, and slightly sunken spots, favored by high humidity and associated with rot (Kimaru et al., 2018). However, the damage caused by *M. velezangeli* originates from its piercing-sucking feeding habit, evidenced by insertion points surrounded by watery areas that subsequently develop into irregular necrosis and can cause deformation or premature fruit drop (Torres Jaimes et al., 2012). In this regard, the shape of the lesion (circular or irregular), the presence of acervuli or punctures, and the distribution of the damage (dispersed or localized) are key to its differentiation.

Fruit damage is associated with oxidation processes and cellular collapse around the feeding site, caused by the injection of salivary enzymes and digestive toxins by phytophagous hemipterans (Backus et al., 2005; Chown & Gaston, 1999). In avocado, the oxidation of compounds such as per-seitol released by damaged tissues contributes to the appearance of necrosis and visible exudates (Hoddle & Hoddle, 2008). In addition, the increase in lesion size over time suggests subsequent processes of enzymatic oxidation, cellular disorganization, and microbial colonization, which accelerate fruit ripening and decay. Similar damage has been reported in cacao and other hosts affected by species of the genus *Monalonion* (Huaycho Callisaya et al., 2017; Laiton et al., 2018).

In shoots and young branches, punctures accompanied by superficial rotting and crystalline exudates indicate alterations in vascular tissues and the release of sugar-rich sap, physiological responses commonly observed in plants attacked by hemipterans (Chown & Gaston, 1999). Symptoms of foliar yellowing and

defoliation suggest interruptions in the transport of water and nutrients, affecting vegetative vigor. Likewise, the high number of punctures per fruit evidences a high individual damage potential and a rapid loss of commercial quality, a pattern reported in Miridae bugs associated with tropical crops (Backus et al., 2005; Huaycho Callisaya et al., 2017).

The morphological characteristics observed in adults, nymphs and eggs are consistent with those described for species of the genus *Monalonion* (Hemiptera: Miridae), particularly with respect to hemelytral coloration, spot patterns, and morphology of the male genitalia. The chromatic variability observed in the forewings, ranging from ochre to black, with the presence of 4–5 reddish spots within the cell near the cuneus, as well as the occasional presence of irregular spots, suggests the existence of intraspecific variation or possible morphotypes within the studied populations (Carvalho & Costa, 1989; Wheeler, 2001). Furthermore, the nymphal development, consisting of five instars, accompanied by a progressive intensification of reddish spots on the body and the presence of antennae with four well-defined segments exhibiting alternating color patterns, is consistent with descriptions of the development of phytophagous mirids associated with tropical crops (Giraldo-Jaramillo et al., 2010; Wheeler, 2001). In addition, the characteristics of the male genitalia, particularly the presence of an aedeagus with abundant membrane, an elongated basal plate, and the asymmetry of the parameres (with the left paramere more developed than the right), constitute taxonomic characters relevant for species delimitation within mirids, since genital structures generally exhibit less intraspecific variability and greater diagnostic value (Carvalho & Costa, 1989; Schuh & Slater, 1995). Taken together, these morphological characteristics support the identification of the analyzed specimens as *Monalonion velezangeli* and are consistent with previous descriptions of populations reported in other countries of the Andean region, reinforcing the morphological characterization of the species and its recognition as an emerging pest in tropical agricultural systems (Giraldo-Jaramillo et al., 2010).

Although the genus *Monalonion* has previously been recorded in Peru (Carvalho & Costa, 1989), the confirmed presence of *Monalonion velezangeli* had not been formally documented. In this study, morphological identification and molecular analysis

using the COI gene allowed confirmation of the species identity, expanding its known distribution in the Andean region, where it has previously been reported in Colombia and Ecuador (Giraldo-Jaramillo et al., 2010; Noboa et al., 2024, 2026).

Phylogenetic analysis based on the mitochondrial COI gene confirmed that all analyzed samples belong to *M. velezungeli*, forming a highly supported monophyletic clade (Huelsenbeck & Ronquist, 2001; Wiley & Lieberman, 2011). The obtained sequences showed a close relationship with records deposited in international databases, suggesting high genetic conservation among regional populations (Hebert et al., 2003; Kim & Jung, 2018). This pattern supports the hypothesis of a recent geographic expansion in Andean agroecosystems. Within the genus, *M. velezungeli* showed greater phylogenetic proximity to *M. bondari*, whereas *M. dissimulatum* was positioned as a more divergent lineage (Folmer et al., 1994; Ratnasingham & Hebert, 2007). Molecular confirmation highlights the importance of integrating genetic tools into phytosanitary monitoring programs, since identification based solely on morphological characters may be complex (Armstrong & Ball, 2005; Raupach et al., 2014; Whaley et al., 2002). Furthermore, the ecological plasticity and the wide host range reported for the species suggest a high adaptive potential, a common characteristic in emerging pests associated with land-use change and agricultural expansion (Suganthi et al., 2023).

The morphological and phylogenetic results obtained in this study contribute to the knowledge of the distribution of *M. velezungeli* in avocado-producing areas, evidencing genetic congruence with populations reported in Colombia and Ecuador, where the species has been widely documented as a pest of avocado (*Persea americana*) and coffee (*Coffea arabica*) (Giraldo-Jaramillo et al., 2010). This suggests that the species may have a broader distribution than currently reported, highlighting the need for multilocus population studies and phytosanitary risk assessments in avocado-producing regions (Ratnasingham & Hebert, 2007).

## Conclusion

This study provides the first confirmed scientific record of *Monalonion velezungeli* affecting avocado

crops in Peru based on morphological and molecular evidence. The presence of this species in avocado-producing areas of the Peruvian Andes expands its known geographic distribution and highlights its potential phytosanitary importance in emerging avocado production systems. The integration of molecular tools such as DNA barcoding proved essential for accurate identification of this pest and may contribute to improved monitoring and early detection of pest expansions. Future studies should address the biology, population dynamics, and host range of this species in order to develop effective monitoring and integrated pest management strategies.

**Author contribution** STL-E. Performed the experiments, analyzed and interpreted the data, data Curation, validated the results, wrote the original draft; VA-R and DB-M. Visualization, performed the experiments; ERR-R and AFH-P. performed the experiments; revised the manuscript, JPA-B. Sequence data validation, funding acquisition, revised the manuscript; JLM-Q validated the results, funding acquisition, revised the manuscript. All the authors approved the final draft for submission to the journal.

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**Data availability** DNA sequences are available in NCBI GenBank (<https://www.ncbi.nlm.nih.gov/genbank/>) under the accession numbers PZ112306, PZ112307, PZ112308 PZ112309, and PZ112310).

## Declarations

**Conflict of interest** The authors declare no competing interests.

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